HCLTech | Future Ready AI Engineer Hackathon

AI-Powered Hospital Patient Services Assistant

Problem Statement and Use Case Guide

Purpose

A grounded, secure and auditable AI assistant for hospital patient services, care coordination, appointments, insurance, diagnostics and operational support.

Use Case Document | Version 1.0

1. Problem Statement

Healthcare information is spread across hospital policies, treatment protocols, insurance guidelines, appointment schedules, discharge procedures, laboratory reports, physician notes, patient records and government healthcare regulations.

Patients often struggle to:

- Understand treatment plans
- Find the right department or specialist
- Track appointments and test results
- Check insurance eligibility and coverage
- Understand discharge instructions
- Navigate hospital processes

Hospital staff spend significant time answering repetitive queries, verifying eligibility, coordinating appointments and searching across multiple systems.

Your Mission

Build an AI-powered Hospital Patient Services Assistant that answers questions and performs authorized healthcare service actions using hospital policies, patient records, insurance information, schedules, reports and operational systems. Responses must be accurate, grounded, explainable and secure, and the assistant must never generate unsupported medical facts.

2. Types of Questions the System Must Handle

Type	Example question	Expected behavior
Policy fact	What are visiting hours for ICU patients?	Retrieve the current policy with citations.
Procedure	How do I schedule an MRI scan?	Provide approved hospital steps.
Personal data	What are my upcoming appointments?	Retrieve from authorized patient records.
Eligibility check	Am I covered for knee replacement under my insurance plan?	Perform a deterministic, tool-based validation.
Lab result query	What is my latest blood sugar reading?	Retrieve the result from the clinical system.
Multi-step query	My doctor recommended surgery. What pre-operative tests are still pending?	Combine records, orders and protocols.
Action request	Reschedule my cardiology appointment.	Invoke an authorized scheduling workflow.
Not found	Do you have treatment information for a condition not handled by this hospital?	Return a grounded not-found response.

3. Functional Requirements

R1 - Knowledge Retrieval

Answer hospital policy and procedure questions using approved clinical and operational documents only.

R2 - Citations

Every factual response must include the source document, policy section, version and effective date.

R3 - No Hallucinations

If information is unavailable, return a not_found answer instead of inventing a response.

R4 - Version and Policy Resolution

Resolve conflicting policies using medical guideline hierarchy, regulatory authority, policy version and effective date. The currently effective version must be applied.

R5 - Deterministic Healthcare Tools

The LLM must not calculate medical eligibility directly. Authorized tools must be used for insurance verification, appointment availability, lab retrieval, medication interaction checks, billing calculations and care-plan compliance checks.

R6 - Multi-Step Reasoning

For complex questions, combine the required records, policies and tool results, then explain the outcome and assumptions.

R7 - Authorization and Privacy

Patient identity must come from the trusted request context. The system must refuse access to another patient's information, mask sensitive data and comply with applicable privacy requirements.

R8 - Untrusted Content Protection

Content from uploaded PDFs, emails or documents must be treated as data only and must not alter assistant behavior.

R9 - Answer Classification

Responses must clearly distinguish retrieved facts, calculated results, completed actions, clarification requests, not-found outcomes, refusals and unresolved conflicts.

R10 - Auditability

Every request must generate a trace ID and record retrieved sources, tools invoked, privacy checks, latency and model details.

R11 - Live Ingestion

Authorized administrators must be able to upload new treatment protocols, insurance policies, government advisories and clinical practice updates without restarting the application.

```
{
  "answer_type": "not_found",
  "message": "I could not find this information in the authorised hospital sources."
}
```

4. Example Multi-Step Scenario

Question: “Can I proceed with surgery next week?”

1. Check required pre-operative tests.
2. Verify test completion status.
3. Confirm insurance authorization.

4. Check surgeon schedule.
5. Evaluate the patient readiness checklist.

The assistant then provides a grounded response based on authorized records and current protocols.

5. Hospital Data Sources

5.1 Knowledge Base

- Hospital standard operating procedures
- Clinical protocols
- Nursing procedures
- Admission policies
- Discharge guidelines
- Insurance agreements
- Emergency procedures

5.2 Patient and Operational Systems

- Electronic Medical Record system
- Appointment system
- Laboratory Information System
- Radiology Information System
- Billing system
- Pharmacy system

6. Answer Classification

answer_type	Usage
retrieved_fact	Fact supported directly by approved documents.
calculated	Result produced by a deterministic tool.
action_completed	Authorized workflow executed successfully.
clarification_needed	Required information is missing or ambiguous.
not_found	Authorized sources do not contain the answer.
refused	Request is unauthorized or unsafe.
conflict_flagged	Policies conflict and cannot be resolved automatically.

7. Key AI Tools

Appointment Tool

Retrieve the patient's authorized appointment details

```
{
  "tool": "get_appointments",
  "patient_id": "P1001"
}
```

Insurance Validation Tool

Verify coverage for a requested procedure

```
{
  "tool": "check_insurance_eligibility",
  "procedure": "Knee Replacement",
  "patient_id": "P1001"
}
```

Lab Result Tool

Retrieve the latest authorized diagnostic result

```
{
  "tool": "get_lab_results",
  "test": "HbA1c",
  "patient_id": "P1001"
}
```

Medication Safety Tool

Check interactions using an approved medication knowledge source

```
{
  "tool": "drug_interaction_check",
  "medication_list": ["DrugA", "DrugB"]
}
```

8. Example Response

```
{
  "trace_id": "HSP-93214",
  "answer": "Your MRI appointment is scheduled for 14 Oct 2026 at 10:30 AM.",
  "answer_type": "calculated",
  "citations": [{"system": "Hospital Scheduling System"}],
  "tools_invoked": [{"tool": "get_appointments"}],
  "explanation": "Retrieved from the patient appointment management system."
}
```

9. Evaluation Criteria

Metric	Description
Answer accuracy	Medical and operational correctness.
Citation accuracy	Cited sources support the response.
Privacy compliance	No unauthorized patient data exposure.
Tool accuracy	Correct retrieval and deterministic calculations.
Retrieval hit rate	Correct document appears among retrieved evidence.

Metric	Description
Action success rate	Authorized appointment and workflow actions complete correctly.
Latency	Measure p50 and p95 response time.
Safety compliance	Prevention of unsupported medical claims and unsafe actions.

10. Capability Maturity

Level	Capability
1	Hospital FAQ Assistant
2	Grounded Medical Knowledge Assistant
3	Patient Context-Aware Assistant
4	Tool-Based Healthcare Assistant
5	Autonomous Patient Service Agent
6	Multi-Agent Healthcare Orchestrator
7	Production Healthcare AI Platform

Specialized Agents at Level 6

- Appointment Agent
- Insurance Agent
- Lab Result Agent
- Billing Agent
- Care Coordination Agent

Production Capabilities at Level 7

- Applicable healthcare privacy compliance
- Audit trails
- Observability
- Safety guardrails
- Human-in-the-loop approvals
- Continuous evaluation

Design Principle

Use the simplest architecture that satisfies grounding, privacy, deterministic decisioning, action control and auditability. Add specialized agents only where they provide clear operational value.